Week 7: SQL Query Implementation for E-Commerce Database

Scenario:

The e-commerce company has collected customer, product, order, and inventory data. The management team wants to retrieve useful information from the database for daily operations and basic business analysis.

1. Perform Basic SQL Queries

Concept:

SQL SELECT command is used to retrieve data from database tables.

1. Display all customer details.

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID CUSTOMER_NAME   EMAIL                PHONE
----------- --------------- -------------------- ----------
          2 Arun            arun@gmail.com       9876543210
          3 Priya           priya@gmail.com      9876543211
          4 Karthik         karthik@gmail.com    9876543212
          5 Divya           divya@gmail.com      9876543213
          6 Rahul           rahul@gmail.com      9876543214
```

2. Display all Product Details

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Laptop
C001             2500             10

P002
Mobile
C001             4500             20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P003
Watch
C004             3400             15

P004
Headphones

PRODUCT_ID
----------
PRODUCT_NAME
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
C001             2500             12

P005
Bag
C002             1500              8
```

3. Retrieve product names and prices only

```
SQL> SELECT Product_Name, Price
  2  FROM Product;

PRODUCT_NAME
----------------------------------------
     PRICE
----------
Laptop
      2500

Mobile
      4500

Watch
      3400


PRODUCT_NAME
----------------------------------------
     PRICE
----------
Headphones
      2500

Bag
      1500
```

4. Display all orders placed by customers.

```
SQL> SELECT O.ORDER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_DATE,
  4         O.TOTAL_AMOUNT,
  5         O.ORDER_STATUS
  6  FROM Orders O
  7  INNER JOIN Customer C
  8  ON O.CUSTOMER_ID = C.CUSTOMER_ID;

  ORDER_ID CUSTOMER_NAME   ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- --------------- --------- ------------ --------------------
      1001 Arun            01-OCT-26         2800 Pending
      1002 Priya           02-OCT-26         4500 Shipped
      1003 Karthik         03-OCT-26         3400 Delivered
      1004 Divya           04-OCT-26         7500 Shipped
      1005 Rahul           05-OCT-26         1500 Pending
```

5. Retrieve payment details.

```
SQL> SELECT * FROM Payment;

PAYMENT_ID     ORDER_ID PAYMENT_D PAYMENT_MODE    PAYMENT_STATUS
------------ ---------- --------- --------------- ---------------
TRANSACTION_AMOUNT
------------------
PAY301             1001 15-OCT-26 UPI             Success
              2800

PAY302             1002 16-OCT-26 Cash            Pending
              4500

PAY303             1003 17-OCT-26 Debit Card      Success
              3400


PAYMENT_ID     ORDER_ID PAYMENT_D PAYMENT_MODE    PAYMENT_STATUS
------------ ---------- --------- --------------- ---------------
TRANSACTION_AMOUNT
------------------
PAY304             1004 18-OCT-26 Credit Card     Success
              7500

PAY305             1005 19-OCT-26 UPI             Success
              1500
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 5000;

no rows selected
```

2. Display products available in stock.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Stock_Quantity > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Laptop
C001             2500             10

P002
Mobile
C001             4500             20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P003
Watch
C004             3400             15

P004
Headphones
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
C001             2500             12

P005
Bag
C002             1500              8
```

ORDER BY Clause

Display products from lowest to highest price.

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

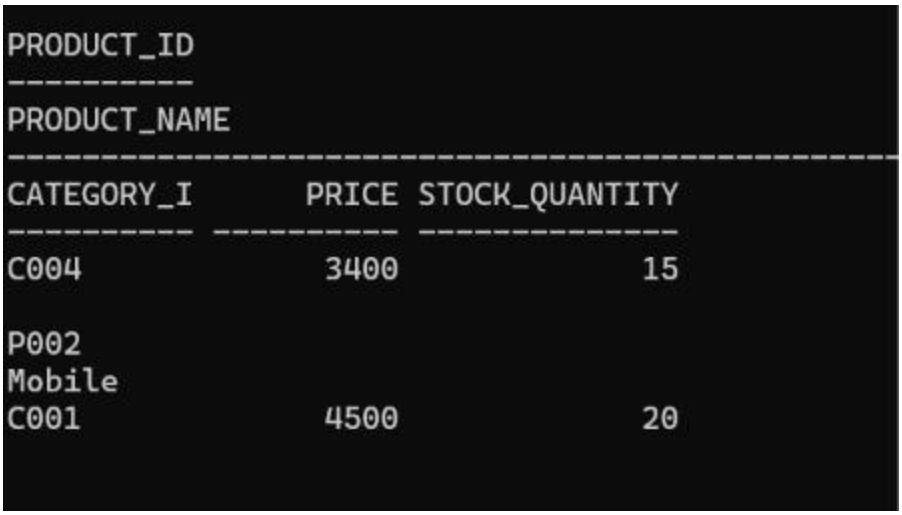
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P005
Bag
C002             1500              8

P001
Laptop
C001             2500             10

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P004
Headphones
C001             2500             12

P003
Watch
```



Find top expensive products.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE ROWNUM <= 5
  4  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
--
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P002
Mobile
C001             4500             20

P003
Watch
C004             3400             15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
--
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P004
Headphones
C001             2500             12
```

```
--------------------------------------------------
--
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- -------------

P004
Headphones
C001             2500             12

P001
Laptop

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
--
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- -------------
C001             2500             10

P005
Bag
C002             1500              8
```

DISTINCT Clause

Display unique product categories.

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_I
----------
C004
C001
C002
```

Find different payment methods.

```
SQL> SELECT DISTINCT PAYMENT_MODE
  2  FROM Payment;

PAYMENT_MODE
---------------
Credit Card
Cash
UPI
Debit Card
```

Search Products Based on Conditions

Find products priced between ₹20,000 and ₹40,000.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 20000 AND 40000;

no rows selected
```

Retrieve Customer and Product Information

Display customer details with their orders.

```
SQL> SELECT Customer.CUSTOMER_ID,
  2         Customer.CUSTOMER_NAME,
  3         Orders.ORDER_ID,
  4         Orders.TOTAL_AMOUNT,
  5         Orders.ORDER_STATUS
  6  FROM Customer
  7  INNER JOIN Orders
  8  ON Customer.CUSTOMER_ID = Orders.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME     ORDER_ID TOTAL_AMOUNT ORDER_STATUS
----------- --------------- ---------- ------------ --------------------
          2 Arun                  1001         2800 Pending
          3 Priya                 1002         4500 Shipped
          4 Karthik               1003         3400 Delivered
          5 Divya                 1004         7500 Shipped
          6 Rahul                 1005         1500 Pending
```

Display product details with category information.

```
SQL> SELECT Product.PRODUCT_ID,
  2         Product.PRODUCT_NAME,
  3         Product.PRICE,
  4         Category.CATEGORY_NAME
  5  FROM Product
  6  INNER JOIN Category
  7  ON Product.CATEGORY_ID = Category.CATEGORY_ID;

PRODUCT_ID   PRODUCT_NAME            PRICE CATEGORY_NAME
---------- -------------------- --------- ----------------
P004         Headphones           2500.00 Electronics
P002         Mobile               4500.00 Electronics
P001         Laptop               2500.00 Electronics
P005         Bag                  1500.00 Clothing
P003         Watch                3400.00 Wearables
```

Multiple Filtering Conditions

Find electronics products costing more than ₹10,000.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE CATEGORY_ID = 'C001'
  4  AND PRICE > 10000;

no rows selected
```

Search products containing the word “Mobile”.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE PRODUCT_NAME LIKE '%Mobile%';

PRODUCT_ID   PRODUCT_NAME         CATEGORY_I     PRICE STOCK_QUANTITY
---------- -------------------- ---------- --------- --------------
P002         Mobile               C001         4500.00             20
```

Business Reports

Product Availability Report

Total Number of Customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM Customer;

TOTAL_CUSTOMERS
---------------
              5
```

Total Orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM Orders;

TOTAL_ORDERS
------------
           5
```

Completed Orders

```
SQL> SELECT *
  2  FROM Orders
  3  WHERE ORDER_STATUS = 'Delivered';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ --------------------
      1003           4 03-OCT-26         3400 Delivered
```

Pending Orders

```
SQL> SELECT *
  2  FROM Orders
  3  WHERE ORDER_STATUS = 'Pending';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ --------------------
      1001           2 01-OCT-26         2800 Pending
      1005           6 05-OCT-26         1500 Pending
```

Most-Reviewed Products

```
SQL> SELECT PRODUCTID, COUNT(*) AS TOTAL_REVIEWS
  2  FROM Review
  3  GROUP BY PRODUCTID
  4  ORDER BY TOTAL_REVIEWS DESC;

PRODUCT TOTAL_REVIEWS
------- -------------
P001                1
P004                1
P002                1
P005                1
P003                1
```

Display Available Products

```
SQL> SELECT *
  2  FROM Product
  3  WHERE STOCK_QUANTITY > 0;

PRODUCT_ID   PRODUCT_NAME         CATEGORY_I     PRICE STOCK_QUANTITY
---------- -------------------- ---------- --------- --------------
P001       Laptop               C001         2500.00             10
P002       Mobile               C001         4500.00             20
P003       Watch                C004         3400.00             15
P004       Headphones           C001         2500.00             12
P005       Bag                  C002         1500.00              8
```

CONCLUSION

Thus, the required SQL operations were performed on the E-Commerce Database using SQL*Plus. The practical demonstrated the use of WHERE, ORDER BY, DISTINCT, BETWEEN, LIKE, JOIN, COUNT, GROUP BY, and CASE statements.